

Quadruped Spider Robot

Complete Assembly, Wiring, and Programming Manual

1. Master Components List (A to Z)

Component	Quantity	Purpose
3D Printed Parts	1 Set (22 pcs)	Body chassis (top/bottom), legs (coxa, femur, tibia).
Arduino Nano V3.0	1	The main brain controlling the servos and logic.
Arduino Nano IO Expansion Shield	1	Provides easy pin access and power distribution for servos.
Battery (7.4V LiPo or 18650 Pack)	1	Primary power source. Needs a step-down buck converter (UBEC) if exceeding 6V.
ESP32-CAM Module	1	Provides live video streaming to the web controller.
FTDI Programmer (CP2102/CH340)	1	Used strictly for uploading code to the ESP32-CAM.
HC-05 Bluetooth Module	1	Receives wireless commands from the controller app.
HC-SR04 Ultrasonic Sensor	1	Detects obstacles for autonomous walking mode.
Jumper Wires (Female-Female)	1 Set	Connecting sensors and modules to the expansion board.
Micro Servos (SG90 or MG90S)	12	3 servos per leg (Coxa, Femur, Tibia) to provide movement.

2. Wiring Workflow & Connections

Follow these steps strictly to ensure the robot moves correctly and doesn't overload.

⚠ **Power Warning:** 12 Servos draw a lot of current. Do NOT power the servos directly from the Arduino Nano's 5V pin. You must supply 5V directly to the Expansion Shield's external power terminal.

Step 1: The 12 Servo Motors

Connect the 12 servos to the digital pins on the Nano Expansion board according to this layout. Ensure the brown wire (GND) faces the outer edge (G or GND pin) of the shield.

- **Leg 0 (Front Left):** Coxa → Pin 3 | Femur → Pin 4 | Tibia → Pin 2
- **Leg 1 (Back Left):** Coxa → Pin 6 | Femur → Pin 7 | Tibia → Pin 5
- **Leg 2 (Front Right):** Coxa → Pin 9 | Femur → Pin 8 | Tibia → Pin 10
- **Leg 3 (Back Right):** Coxa → Pin 12 | Femur → Pin 11 | Tibia → Pin 13

Step 2: HC-05 Bluetooth Module

- **VCC** → 5V on Expansion Board
- **GND** → GND on Expansion Board
- **TXD** → RX (Pin D0) on Arduino Nano
- **RXD** → TX (Pin D1) on Arduino Nano
- *Note: Unplug the RX/TX pins when uploading code to the Arduino via USB!*

Step 3: HC-SR04 Ultrasonic Sensor

- **VCC** → 5V
- **GND** → GND
- **Trig** → Analog Pin A0
- **Echo** → Analog Pin A1

Step 4: ESP32-CAM Module

- **5V** → 5V on Expansion Board
- **GND** → GND on Expansion Board
- *(No data wires are connected to the Arduino. The camera runs independently and streams to the network.)*

3. Arduino Nano Code Upload Workflow

There are multiple code files provided in the `Arduino_Codes` folder. Here is the order of operations:

1. **Mechanical Assembly Phase (Upload `3_basic_position.ino`):**

Before attaching the legs to the servo horns, upload this code. It locks all 12 servos to exactly 90 degrees. Attach your robot legs so they are perfectly straight and perpendicular while this code is running.

2. **Testing Phase (Upload `1_servo_test.ino`):**

Upload this to verify every servo sweeps smoothly from 0 to 180 degrees. Ensure no legs get physically stuck or blocked.

3. **Final Operational Phase (Upload `4_repetitive_movements.ino`):**

This is the final master code. It contains the Inverse Kinematics, Bluetooth command listening, and Obstacle Avoidance logic. Upload this for final use with the Web App Controller.

4. ESP32-CAM Coding & Finding the IP Address

To stream video to your controller app, you must flash the ESP32-CAM using an FTDI programmer.

Step A: FTDI to ESP32-CAM Wiring for Uploading

- **5V** → 5V
- **GND** → GND
- **TX** → U0RXR
- **RX** → U0TXD
- **IO0** → GND (*Critical: This puts the board in flash mode*)

Step B: Arduino IDE Setup

1. Open Arduino IDE. Go to **File > Examples > ESP32 > Camera > CameraWebServer**.
2. In the **Tools** menu:
 - Board: **AI Thinker ESP32-CAM**
 - Upload Speed: **115200**

Step C: Code Modifications

In the CameraWebServer sketch, make the following changes:

```
// 1. Uncomment the AI Thinker camera model, comment out the rest #define  
CAMERA_MODEL_AI_THINKER // 2. Insert your WiFi Credentials const char* ssid =  
"YOUR_WIFI_NETWORK_NAME"; const char* password = "YOUR_WIFI_PASSWORD";
```

Step D: Upload and Get IP

1. Click **Upload**. Wait for the "Writing... 100%" message.
2. **Remove the jumper wire connecting IO0 to GND.**
3. Press the RESET button on the ESP32-CAM (or cycle power).
4. Open the Arduino **Serial Monitor** (Set baud rate to 115200).
5. You will see text output connecting to WiFi, followed by:

```
Camera Ready! Use 'http://192.168.1.xxx' to connect
```

6. Take that IP address, append :81/stream to it (e.g., **http://192.168.1.100:81/stream**), and paste it into the Camera tab of your Web Controller App.